

Wuqian Tang

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Education

National Tsing Hua University

Hsinchu, Taiwan

Ph.D. Candidate in Computer Science (GPA: 4.25/4.3)

Sep. 2023 – Jul. 2027 (Expected)

Supervisors: Prof. Chun-Yao Wang and Prof. Bei Yu (The Chinese University of Hong Kong)

National Tsing Hua University

Hsinchu, Taiwan

M.S. in Computer Science (GPA: 4.11/4.3)

Sep. 2021 – Jul. 2023

Supervisor: Prof. Chun-Yao Wang

Ming Chuan University

Taoyuan, Taiwan

B.S. in Computer Science and Information Engineering

Sep. 2017 – Jun. 2021

Selected Awards and Honors

Second Place in MLCAD 2026 Contest

Sep. 2026

Excellent Teaching Assistant Award, National Tsing Hua University

Aug. 2026

Outstanding Teaching Assistant Award (3×), National Tsing Hua University

Aug. 2025, Feb. 2026, Aug. 2026

Distinguished Alumnus Award, Department of CSIE, Ming Chuan University

Jun. 2026

2nd Place in IWLS 2026 Programming Contest

May 2026

Hsinchu City Outstanding Youth (College/University Division)

Mar. 2026

1st Place Award (Prob. C) & 2nd Place Award (Prob. A) in ICCAD 2025 CAD Contest

Oct. 2025

First Place in ICCAD 2025 CADathlon Programming Contest

Oct. 2025

Presidential Hsing Chien Award, NTHU (1 of 13 university-wide recipients for holistic excellence)

May 2025

2nd Place Award (Prob. A) & Honorable Mention (Prob. B) in ICCAD 2024 CAD Contest

Oct. 2024

Second Place in ICCAD 2024 CADathlon Programming Contest

Oct. 2024

Synopsys - Prof. Chung Laung Liu Ph.D. Scholarship

Sep. 2024 – Present

“Qilin Cup” (1st among 400+ teams) in Integrated Circuit EDA Elite Challenge

Dec. 2023

Honorable Mention (Prob. A) in ICCAD 2023 CAD Contest

Oct. 2023

Third Place in 2023 EDathon

Aug. 2023

1st Place Award (Prob. A) in ICCAD 2022 CAD Contest

Oct. 2022

National Tsing Hua University Mainland China Student Scholarship

Sep. 2021 – Present

Silver Award in International Collegiate Programming Contest (ICPC Regional)

Nov. 2020

Work Experience

Giga Design Automation

Shanghai, Mainland China

R&D Intern in Logic Synthesis Group (SYN), PI: Dr. Yun Shao

Jun. 2025 – Sep. 2025

Shenzhen Research Institute, The Chinese University of Hong Kong

Shenzhen, Mainland China

Assistant Researcher in Design Automation Laboratory, PI: Prof. Bei Yu

Jan. 2024 – Mar. 2024

Teaching Experience

Teaching Assistant, National Tsing Hua University

Hsinchu, Taiwan

- Introduction to Programming (Prof. Yu-Guang Chen, Department of Electrical Engineering) *Fall 2026*
- Hardware Security (Prof. Yu-Guang Chen and Prof. TingTing Hwang, Department of CS) *Spring 2026*
- Advanced Logic Synthesis (Prof. TingTing Hwang, Department of Computer Science) *Spring 2025, Spring 2026*
- Introduction to Integrated Circuit Design (Prof. Yu-Guang Chen, Department of Computer Science) *Fall 2025*
- Computer Architecture (Prof. TingTing Hwang, Department of Computer Science) *Fall 2024, Fall 2025*
- Design Automation (I) (Prof. Jeong-Tyng Li, College of Semiconductor Research) *Spring 2025*

Technical Skills

Programming: Verilog, C/C++, CUDA, Python, Java, SQL, and MATLAB

Certification: Expert Level in Collegiate Programming Examination (certified by ACM-ICPC Taiwan Council)

Language: Fluent in English, Native in Mandarin (Proficient in Traditional and Simplified Chinese)

Publications

An asterisk (*) indicates equal contribution.

[J1] Fangzhou Liu, **Wuqian Tang**, Zehua Pei, Ziyang Yu, Haisheng Zheng, Zhuolun He, Mengjia Dai, Tinghuan Chen, Bei Yu, “CB-EVO: Contextual Bandit Tuning with Evolutionary Search for Logic Synthesis”, *ACM Transactions on Design Automation of Electronic Systems (TODAES)*, vol. 31, no. 6, pp. 119:1–119:26, 2026.

[J2] Yuh-Pyng Shieh, Shao-Hang Hsu, **Wuqian Tang**, Wei-Ying Ke, Po-sheng Ou, Yu-wei Chang, Ming-chorng Hwang, “The Lunar-Month-Interval Algorithm for Dating Oracle Bone Inscriptions”, *Journal of Digital Archives and Digital Humanities (DADH)*, vol. 15, pp. 1–35, Apr. 2025 (in Chinese).

[C19] Yu-Cheng He, Han-Wen Tsao, Yung-Chih Chen, Yi-Ting Li, **Wuqian Tang**, Jian-Meng Yang, Chun-Yao Wang, “Observability-Aware Pattern Generation for Hardware Trojan Detection”, *IEEE International Conference on Computer Design (ICCD)*, Hong Kong SAR, China, Nov. 16–18, 2026 (*to appear*).

[C18] Bo-Ying Wang, Yong-Fong Chang, Hsun-Wei Pao, Yung-Chih Chen, Tien-Hsin Peng, Yu-Ting Hung, Tsai-Chieh Cheng, Yi-Ting Li, **Wuqian Tang**, Chun-Yao Wang, Shih-Chieh Chang, “A Design Rule-Aware Post-Route ECO Framework for Fixing Electromigration Violations”, *ACM/IEEE International Conference on Computer-Aided Design (ICCAD)*, San Jose, California, USA, Nov. 8–12, 2026 (*to appear*).

[C17] **Wuqian Tang***, Zixiao Wang*, Che-Chien Lin, Fong-Hua Fu, Laura Angélica Shion-Korsak, Xinyun Zhang, Bei Yu, Chun-Yao Wang, “TAPCO: LLM-Guided Design-Adaptive Timing and Power Co-Optimization”, *ACM/IEEE International Symposium on Machine Learning for CAD (MLCAD)*, Jeju, South Korea, Sep. 7–9, 2026.

[C16] Tao-Chun Huang, Hao-Yu Tsai, You-Cheng Lin, Yi-Ting Li, **Wuqian Tang**, Yung-Chih Chen, Jian-Meng Yang, Chun-Yao Wang, “Late Breaking Results: QUBO Approaches to Logic Equivalence Checking and Testing”, *ACM/IEEE Design Automation Conference (DAC)*, Long Beach, California, USA, Jul. 26–29, 2026.

[C15] Fangzhou Liu*, **Wuqian Tang***, Bo-Ying Wang, An-Chieh Shen, Han-Wen Tsao, Yuyang Ye, Yun Shao, Chun-Yao Wang, Bei Yu, “HyPAS: A Hybrid Optimization Framework for Placement and Sizing Co-Optimization”, *ACM/IEEE Design Automation Conference (DAC)*, Long Beach, California, USA, Jul. 26–29, 2026.

[C14] Chi-Wei Chen, Yi-Ting Li, **Wuqian Tang**, Yung-Chih Chen, Jian-Meng Yang, Chun-Yao Wang, “Learning to Approximate: Circuit Learning and Deep Reinforcement Learning for Approximate Logic Synthesis with an Error Rate Guarantee”, *IEEE/ACM Design, Automation and Test in Europe (DATE)*, Verona, Italy, Apr. 20–22, 2026.

[C13] You-Cheng Lin, Yi-Ting Li, **Wuqian Tang**, Yung-Chih Chen, Chia-Chieh Chu, Chun-Yao Wang, “A Mathematical Exploration to Equivalence Checking of Quantum Circuits”, *IEEE/ACM Design, Automation and Test in Europe (DATE)*, Verona, Italy, Apr. 20–22, 2026.

- [C12] Zi-Wei Huang, Yi-Ting Li, **Wuqian Tang**, Liang-Chia Cheng, Yung-Chih Chen, Chun-Yao Wang, “Process Selection Strategy for Chiplet-Based 2.5D Systems under Design Constraints”, *IEEE International VLSI Symposium on Technology, Systems and Applications (VLSI-TSA)*, Hsinchu, Taiwan, Apr. 13–17, 2026.
- [C11] Yi-Ting Shen, Yan-Hsiu Liu, Yi-Ting Li, **Wuqian Tang**, Yung-Chih Chen, Hao-Chiang Shao, Chia-Wen Lin, Chun-Yao Wang, “Integrating Automatic Prompt Engineering and Vision-Language Model for Pad Defect Classification”, *IEEE International Symposium on Quality Electronic Design (ISQED)*, San Francisco, California, USA, Apr. 8–10, 2026.
- [C10] Kai-Po Hsu, Yi-Ting Li, Fu-Cheng Cai, Chia-Feng Chien, **Wuqian Tang**, Ting-Yu Ku, Yu-Chen Cheng, Tsung-Han Lai, Cheng-Lung Wang, Yi-Ting Shen, Kuan-Ling Chou, Zi-Wei Huang, Tao-Chun Huang, Tzu-Li Hsu, Yung-Chih Chen, Shih-Chieh Chang, Ting-Chi Wang, TingTing Hwang, Chun-Yao Wang, “An Efficient Approach to NP3-Boolean Matching: Leveraging Multi-bit Ports and Reverse Engineering to Minimize Search Space”, *IEEE International Symposium on Quality Electronic Design (ISQED)*, San Francisco, California, USA, Apr. 8–10, 2026.
- [C9] Yu-Chuan Yen, Fu-Cheng Cai, Yi-Ting Li, **Wuqian Tang**, Yung-Chih Chen, Ihao Chen, Chun-Yao Wang, “Area-Oriented Threshold Logic Circuit Synthesis Using Negative Weights”, *IEEE International Symposium on Quality Electronic Design (ISQED)*, San Francisco, California, USA, Apr. 8–10, 2026.
- [C8] Chia-Feng Chien, Ming-Wei Chen, Yin-Chen Chang, Yung-Chih Chen, Yan-Hsiu Liu, Hao-Chiang Shao, Chia-Wen Lin, Yi-Ting Li, **Wuqian Tang**, Chun-Yao Wang, “Improving CNN-based Pad Defect Classification with Enhanced Preprocessing Techniques”, *IEEE International SoC Design Conference (ISOCC)*, Busan, South Korea, Oct. 15–18, 2025.
- [C7] Kuan-Ling Chou, Cheng-Lung Wang, Yung-Chih Chen, **Wuqian Tang**, Yi-Ting Li, Shih-Chieh Chang, Chun-Yao Wang, “CNN Model Optimization Using a Hybrid Approach of Genetic Algorithm-based Pruning and Retraining with Knowledge Distillation”, *ACM Great Lakes Symposium on VLSI (GLSVLSI)*, New Orleans, Louisiana, USA, Jun. 30 – Jul. 2, 2025.
- [C6] Yu-Che Lee, Yu-Hsuan Chen, Yu-Chen Cheng, Yong-Fong Chang, Jia Wei Lin, Hsun-Wei Pao, Yung-Chih Chen, Yi-Ting Li, **Wuqian Tang**, Shih-Chieh Chang, Chun-Yao Wang, “Real-Time Dynamic IR-drop Prediction for IR ECO”, *ACM/IEEE Design Automation Conference (DAC)*, San Francisco, California, USA, Jun. 22–25, 2025.
- [C5] Kai-Po Hsu, Tsung-Han Lai, Yi-Ting Li, **Wuqian Tang**, Yun-Ju Lee, Yung-Chih Chen, Wen-Hsin Chiu, Chun-Yao Wang, “IMU-based Motion Trajectory Reconstruction and Recognition with Dynamic Calibration on Embedded System Platform”, *IEEE International Symposium on Quality Electronic Design (ISQED)*, San Francisco, California, USA, Apr. 23–25, 2025.
- [C4] Yu-Cheng Lee, Kai-Po Hsu, Yun-Ju Lee, Yi-Ting Li, **Wuqian Tang**, Yung-Chih Chen, Wen-Hsin Chiu, Chun-Yao Wang, “On Construction of Trajectory of Boxer’s Punch using a single IMU”, *IEEE International Symposium on Intelligent Signal Processing and Communication Systems (ISPACS)*, Kaohsiung, Taiwan, Dec. 10–13, 2024.
- [C3] **Wuqian Tang**, Chuan-Shun Huang, Yung-Chih Chen, Yi-Ting Li, Shih-Chieh Chang, Chun-Yao Wang, “Model Reduction Using a Hybrid Approach of Genetic Algorithm and Rule-based Method”, *IEEE International System-on-Chip Conference (SOCC)*, Dresden, Germany, Sep. 16–19, 2024.
- [C2] Chuan-Shun Huang, **Wuqian Tang**, Yung-Chih Chen, Yi-Ting Li, Shih-Chieh Chang, Chun-Yao Wang, “An Efficient Approach to Iterative Network Pruning”, *IEEE International VLSI Symposium on Technology, Systems and Applications (VLSI-TSA)*, Hsinchu, Taiwan, Apr. 22–25, 2024.
- [C1] **Wuqian Tang**, Yi-Ting Li, Kai-Po Hsu, Kuan-Ling Chou, You-Cheng Lin, Chia-Feng Chien, Tzu-Li Hsu, Yung-Chih Chen, Ting-Chi Wang, Shih-Chieh Chang, TingTing Hwang, Chun-Yao Wang, “A Hybrid Approach to Reverse Engineering on Combinational Circuits”, *IEEE/ACM Design, Automation and Test in Europe (DATE)*, Valencia, Spain, Mar. 25–27, 2024.